

AMQUI, QC, Canada

WMO: 713860

Lat: 48.472N

Lon: 67.435W

Elev: 166

StdP: 99.35

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.8	-24.2	-30.2	0.2	-26.4	-27.8	0.3	-23.8	10.4	-10.9	9.6	-11.6	2.2	170	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.3	19.2	26.6	18.3	24.9	17.7	21.1	25.7	20.1	24.1	19.1	22.9	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.6	23.0	18.6	13.7	22.1	17.6	12.9	21.1	62.1	25.9	58.4	24.2	55.1	23.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.0	-31.9	31.4	3.2	1.5	-34.2	32.5	-36.1	33.4	-37.9	34.2	-40.2	35.3		
(5)			-31.7	23.0	2.7	1.3	-33.6	24.0	-35.2	24.7	-36.6	25.5	-38.6	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-12.6	-11.2	-5.8	1.8	9.4	14.7	18.1	17.2	12.6	5.9	-0.7	-7.8
	DBStd	11.67	6.66	6.29	6.12	4.11	4.07	3.70	2.82	3.01	4.03	4.11	4.84	5.91
	HDD10.0	3127	699	594	490	247	61	4	0	0	18	139	322	552
	HDD18.3	5467	958	828	748	496	278	120	38	58	178	384	571	811
	CDD10.0	774	0	0	0	1	43	145	252	223	96	13	1	0
	CDD18.3	73	0	0	0	0	1	11	32	23	5	0	0	0
	CDH23.3	717	0	0	0	0	47	139	277	198	53	2	0	0
	CDH26.7	142	0	0	0	0	9	32	53	38	9	0	0	0
Wind	WSAvg	3.0	3.3	3.3	3.3	3.2	3.0	2.6	2.3	2.3	2.6	3.0	3.2	3.3
Precipitation	PrecAvg	1062	78	72	75	68	84	92	108	97	97	100	96	94
	PrecMax	1195	154	109	131	119	137	164	179	179	198	189	195	152
	PrecMin	860	14	28	40	24	31	25	26	20	32	27	21	45
	PrecStd	97	34	24	28	25	29	31	32	39	39	36	35	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	4.9	10.6	18.4	27.7	30.2	30.3	30.0	27.9	22.0	15.8	6.6
		MCWB	4.6	2.6	5.4	10.5	15.1	19.8	20.6	20.8	20.4	15.3	14.0	5.6
	2%	DB	2.5	2.5	7.1	13.7	24.0	26.9	28.1	27.6	24.7	17.6	10.7	3.8
		MCWB	1.6	1.2	3.1	7.1	14.2	17.3	19.3	19.6	18.1	13.7	8.8	2.8
	5%	DB	0.3	0.7	4.7	10.6	20.6	24.6	26.4	25.5	22.1	14.9	7.9	1.8
		MCWB	-0.5	-0.4	1.9	5.5	12.6	16.4	18.7	18.7	16.9	11.3	6.2	0.9
	10%	DB	-2.5	-1.3	2.8	8.3	17.6	22.3	24.6	23.6	19.7	12.4	5.7	0.5
		MCWB	-3.4	-2.4	0.6	4.3	11.0	15.6	18.0	17.7	15.7	10.0	4.2	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	3.3	6.7	11.8	18.2	21.6	22.8	22.1	21.6	16.6	13.8	5.8
		MCDB	5.2	4.0	8.8	16.1	23.4	27.5	26.7	27.5	26.1	19.6	15.6	6.3
	2%	WB	1.8	1.2	3.7	8.1	15.5	19.6	21.2	20.9	19.4	14.1	9.2	2.9
		MCDB	2.3	1.9	5.9	12.3	20.9	24.1	25.4	25.6	22.8	17.0	10.3	3.6
	5%	WB	-0.6	-0.6	2.2	6.2	13.6	18.1	20.1	19.7	17.9	12.1	6.3	1.1
		MCDB	0.3	0.5	4.5	9.7	18.9	22.4	24.2	23.6	21.3	14.4	7.4	1.7
	10%	WB	-3.4	-2.4	0.8	4.7	11.9	16.7	19.1	18.7	16.2	10.2	4.3	-0.3
		MCDB	-2.6	-1.4	2.7	7.8	16.7	20.6	23.0	22.2	19.2	12.4	5.6	0.5
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	10.7	9.4	11.3	11.2	10.4	10.6	10.4	8.1	6.4	7.4
		MCDBR	10.6	10.2	12.4	14.5	17.8	15.8	13.9	13.7	14.2	12.7	8.9	7.3
		MCWBR	10.3	8.8	9.2	8.8	9.1	7.5	6.6	7.0	8.4	8.6	7.5	7.1
	5% WB	MCDBR	10.5	8.7	10.4	12.5	14.4	12.2	11.0	11.5	12.2	11.0	8.2	7.5
		MCWBR	10.3	7.9	8.2	8.5	8.2	6.8	6.2	6.5	8.0	8.3	7.5	7.4
Clear-Sky Solar Irradiance	taub	0.255	0.271	0.301	0.361	0.355	0.373	0.400	0.379	0.356	0.327	0.292	0.274	
	taud	2.315	2.266	2.229	2.258	2.409	2.376	2.322	2.383	2.436	2.505	2.498	2.321	
	Ebn at Noon	824	890	914	887	906	888	856	860	848	820	782	743	
	Edh at Noon	70	93	115	125	112	117	121	109	94	75	60	62	
All-Sky Solar Radiation	RadAvg	1.24	2.12	3.33	4.36	5.05	5.46	5.29	4.58	3.48	2.04	1.06	0.81	
	RadStd	0.14	0.24	0.24	0.37	0.39	0.39	0.56	0.31	0.26	0.18	0.14	0.08	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (22 neighbors)		N/A	N/A	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page